IV - Plant Taxonomy - Flower Dissection

Aim : To study, identify, dissect and describe flowering plants of families Fabaceae, Apocynaceae, Solanaceae, Euphorbiaceae and Musaceae.

Principle: Taxonomy deals with identification, nomenclature and classification of organisms. Bentham and Hooker's system of classification is universally used for classification of plants. Field identification of plants is based primarily on morphological features particularly the floral characters.

Requirements: Locally available plant specimens of Clitoria ternatea, Catharanthus roseus, Datura metel, Ricinus communis, Musa paradisiaca. Each specimen should have at least a small branch with a few internodes, leaves, flowers and fruits; glass slides, cover glass, petridish, blade, needles, brush, hand lens, dissecting microscope and compound microscope.

Exercise: 23

Fabaceae – *Clitoria ternatea*

Systematic position

Kingdom : Plantae
Clade : Angiosperms
Clade : Eudicots
Clade : Rosids
Order : Fabales
Family : Fabaceae

Floral characters:

Inflorescence : Solitary and axillary cyme.

Flower : Bractate, bracteolate, bisexual, zygomorphic, pentamerous and hypogynous.

Calyx : Sepals 5, synsepalous, Valvate aestivation, odd sepal is anterior in position.

Corolla : Petals 5, apopetalous, Papilonaceous corolla and descendingly imbricate.

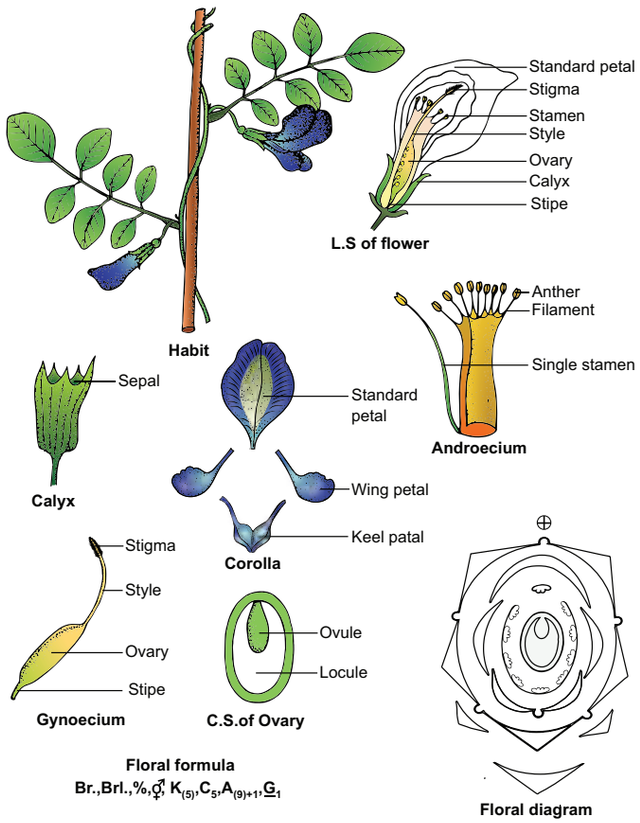


Figure 23 : *Clitoria ternatea*

Androecium : Stamens 10, diadelphous, (9) + 1.

Gynoecium: Monocarpellary, unilocular and ovules on marginal placentation, Superior ovary.

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